

Mysticeti is Sui’s production BFT protocol: an **uncertified block DAG** that commits in **three message rounds** in the good case, with a **fast path woven into the same DAG** for owned-object txs. It replaced Narwhal + Bullshark on Sui mainnet (July 2024) and was tightened again as **Mysticeti v2**.

What it replaced

Narwhal built a **certified DAG**: each block needed a quorum certificate before it was a first-class vertex. That gave availability and equivocation resistance, but:

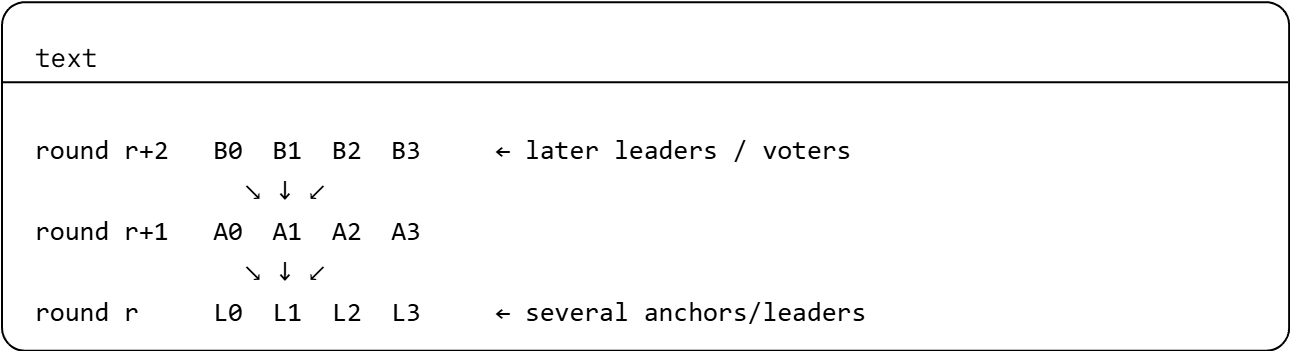
- ~3 extra rounds of signing **per block** just to certify
- $O(n)$ signature work per block
- Bullshark then ran a slower commit rule on that certified graph

Result on Sui: shared-object commits around **~1.9 s**, high validator CPU. Mysticeti’s bet: **drop explicit certificates** and read the same properties off **DAG patterns**. Reported mainnet drop: **~1.9 s → ~400 ms** consensus, with much less CPU.

Model

- $n = 3f + 1$ validators, f Byzantine
- Time is **rounds**, not slots of a single leader
- Every honest validator proposes a block each round
- A block carries: epoch, round, author, timestamp, $\geq 2f + 1$ **ancestor refs**, transactions, commit votes

Progress rule: advance to round r after seeing a quorum of round $r - 1$ blocks (threshold logical clock). The DAG is the clock and the vote store at once.



Many proposers per round ⇒ bandwidth from all links, not one leader’s NIC. Censorship resistance is structural: your tx can sit in any honest block

Resistance is structural, your ex can't fix it any honest block.

Uncertified DAG

A block is **one author signature**, then gossiped. No QC per vertex.

Quorum support is inferred:

- **Certificate pattern:** $2f + 1$ later blocks in the causal future **support** (link to) a proposal
- **Skip pattern:** $2f + 1$ later blocks **do not** support it

Those patterns *are* the certificates. Safety still requires that two conflicting commits cannot both collect $2f + 1$ honest support; availability of committed data follows from the fact that committed anchors sit in the causal history of later honest blocks (plus a synchronizer — Beluga — when the network is ugly).

CPU win: one sign/verify per **block**, not per block plus n cert signatures, and not per transaction.

Commit rule (Mysticeti-C)

Each round designates **one or more anchors** (leaders). The commit rule walks the DAG and either **commits** or **skips** each anchor.

- **Direct commit:** enough support appears within the 3-round window so the anchor is decided immediately.
- **Indirect commit:** a later directly committed anchor (round $\geq R + 3$) has certificates pointing at an earlier undecided anchor; that earlier one commits in the later anchor's shadow.

Once an anchor commits, every still-uncommitted block in its causal history is linearized into the **commit sequence**. That sequence is the total order used for **shared objects**.

Design goals vs Bullshark:

- Commit **every round**, not every other round
- **Several leaders per round** so a crashed leader does not stall the pipeline (no classic head-of-line)
- Steady-state latency = **3 message delays**, the PBFT / HotStuff good-case lower bound for BFT agreement

Paper / lab numbers (not mainnet SLA): **~0.5 s** commit at **~200k TPS**; 10-node tests held

300k TPS before 1 s latency. Production consensus latency is advertised in the **~300–400 ms** range depending on version and region.

After a commit, Sui still **schedules execution by shared-object version**, and can **cancel** txs after consensus if a commit is too fat — consensus throughput and execution capacity are deliberately decoupled.

Fast path (Mysticeti-FPC)

Owned-object txs do not need a global order versus the whole chain. They need: **no conflicting mutate of the same owned object**, and **availability**.

FPC **does not run a second protocol**. Votes are **woven into DAG blocks**:

- A validator includes a fast-path tx in a block only if it does not conflict with txs it already voted for
- Other validators' later blocks that link that history are implicit votes
- $2f + 1$ votes $\Rightarrow$ the tx is **certified** and can **execute at the end of round 2** (one round sooner than full C commit)

Checkpoints later include certified fast-path txs that appear in the causal history of C commits, so the two paths reconverge for state snapshots. Signature cost stays per-block, unlike FastPay/Zef-style per-tx certs.

That is the consensus half of “Sui owned objects skip full consensus.”

Mysticeti v2

v1 still had a **pre-consensus validation / Quorum Driver** path: blast tx to many validators, collect signatures, then enter the DAG. v2 folds validation into the DAG:

- **Accept is implicit** (you appear as an ancestor)
- **Reject is explicit** (only then extra votes)
- Certification of a tx is evaluated when needed, across multiple rounds so a tx submitted via a slow validator can still finalize
- **Transaction Driver** replaces Quorum Driver: send to **one** well-chosen validator; that node inserts into consensus and returns certified effects; retry elsewhere on failure

Reported fullnode latency cuts on the order of **25–35%** depending on region. Conceptually v2 applies the **leader commit rule to transactions concurrently** with block commit/skip.

How this hooks to Sui execution

text

owned-only tx → FPC / v2 implicit cert → execute immediately
(object versions already unique by owner)

shared-object tx → C commit linearizes block
→ consensus assigns shared object versions
→ execute txs that write obj @ v then @ v+1 in order
→ different objects still parallel

Mysticeti does **not** replace Block-STM. It **produces the order (and sometimes skips ordering)**. Execution is a separate, object-version scheduler. Aptos: BFT block then STM. Sui: DAG decides *whether* a total order is required, then a much simpler executor.

Properties and caveats

Strengths

- Good-case latency at the 3-round bound
- All-to-all proposing → throughput and censorship resistance
- Crash of one leader masked by other anchors
- Fast path and consensus share one message fabric
- Cheap crypto relative to certified DAGs

Limits

- **Liveness is sensitive to round-jumping.** If honest nodes skip rounds arbitrarily, there is a published counterexample where nothing ever commits. Mechanized proofs (IEEE S&P 2026) restore liveness with a restriction on jumping. Implementations must not “optimistically” leap rounds.
- **Ordering is not MEV-proof.** Parallel proposers and post-commit fee sort still leave insertion / delay / bias surface; that is an active research critique, not a solved feature.
- Shared-object hotspots still queue; consensus being fast does not make one AMM object infinitely parallel.
- Uncertified blocks need a real **block sync** path when links drop (Beluga).

Compared with Aptos consensus (for the earlier thread)

	Mysticeti (Sui)	AptosBFT + Block-STM
Structure	Uncertified multi-leader DAG	Pipelined leader BFT + ordered blocks
What is agreed	Anchor commits + object versions / fast-path certs	A total order of all user txs
Fast path	First-class (FPC / v2)	None at consensus; STM only speeds execution
Latency bound	3 rounds (C), ~2 for owned (FPC)	Quorum votes on blocks; then execute
Execution coupling	Object graph	STM over the whole block

Mysticeti is the reason Sui can say “owned transfer does not wait on the DEX.” Block-STM is the reason Aptos can say “write normal Move, we will sort it out in the block.” They sit at different layers: Mysticeti is specifically the **ordering substrate for an object-centric chain**.